

Algorithm for file updates in Python

Project description

The main objective of the project is to automate the process of maintaining an allow list for updating the IP addresses that are allowed to access the healthcare subnetwork which is restricted. The allow list has the IP addresses of those employees who are allowed to access the restricted patient record information, whereas the remove list has those IP addresses that are not allowed to access this information anymore. This process was accomplished through Python by reading, converting, comparing, and writing the information back in the file.

Open the file that contains the allow list

The filename should be assigned to the variable `import_file`, which is a string, as follows: `import_file = "allow_list.txt"`. The `with` statement and the `open()` method are used to open the file, and then the opened file is assigned to the variable `file`. By opening the file in the read mode, it becomes safe to access its content within the `with` block.

Code:

```
import_file = "allow_list.txt"
with open(import_file, "r") as file:
```

Read the file contents

Within the `with` block use the `.read()` method on the `file` object to retrieve the content of the allow list as a single string. Assign the returned string to `ip_addresses`.

Code:

```
ip_addresses = file.read()
```

Convert the string into a list

Use the `.split()` method on the `ip_addresses` string to separate the IP addresses into individual list elements. Assign the resulting list back to `ip_addresses`. Because the IP addresses are separated by whitespace/newlines in the file, `.split()` converts the string into a list that can be modified.

Code:

```
ip_addresses = ip_addresses.split()
```

Iterate through the remove list

Use the `for` keyword to create a loop that goes through each item in `remove_list`. The loop variable `element` represents the current IP address being checked during each iteration.

Code:

```
for element in remove_list:
```

Remove IP addresses that are on the remove list

Within the `for` loop, use an `if` statement with the `in` operator to check whether `element` is present in the `ip_addresses` list. If it is present, use the `.remove()` method to delete that IP address from the allow list.

Code:

```
if element in ip_addresses:  
    ip_addresses.remove(element)
```

Applying `.remove()` this way is possible because there are no duplicate IP addresses in the `ip_addresses` list, so each matching IP address only needs to be removed once.

Update the file with the revised list of IP addresses

First, use the `.join()` method on the string `"\n"` to convert the revised `ip_addresses` list back into a single string, placing each IP address on a separate line. Then use another `with` statement and `open()` to reopen the file in write mode (`"w"`). The `.write()` method writes the revised string over the existing contents of the allow list file.

Code:

```
ip_addresses = "\n".join(ip_addresses)  
with open(import_file, "w") as file:  
    file.write(ip_addresses)
```

Summary

Algorithm starts with opening the `allow_list.txt` file and storing its contents in a string. String is then made into a list because we must remove certain IPs from that list one at a time. For this purpose, a `for` loop iterates through all the IPs stored in `remove_list` and if any of these IPs are found in the allow list, the IP is removed using the function `.remove()`. List is once again transformed into a string with new lines using `.join()` and stored in the original file.